

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0048

Dresden Generating Station 230 kV
March 03, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0048. The Interconnection Customer proposes a 513.3 MW natural gas combined cycle facility interconnecting at the Dresden Generating Station 230 kV substation in the Illinois area. The request was received on June 19, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 41 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,371,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Morris Cogeneration Llc-Dresden Generating Sta 230 kV L is overloaded to 115.2% of its 599 MVA emergency rating for the loss of the Morris Cogeneration Llc-Boca Raton 230 kV. The project contributes 27.2% DFAX (139.6 MW) to this overload.

For the loss of Brown Plant-Morris Cogeneration Llc 230 kV Ckt 1, the Brown Plant-Morris Cogeneration Llc 230 kV loads to 108.6% of its 696 MVA rating. Project DFAX: 19.4%.

For the loss of Morris Cogeneration Llc-Brown Plant 230 kV Circuit 2, the Brown Plant-Morris Cogeneration Llc 230 kV Ckt 1 loads to 100.7% of its 750 MVA rating. Project DFAX: 19.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Morris Cogeneration Llc-Dresden Generating Sta 230 kV L	Rebuild line with higher-capacity conductor	\$9,758,000	74% (\$7,217,000)
Brown Plant-Morris Cogeneration Llc 230 kV	Reconductor line	\$9,248,000	68% (\$6,315,000)
Morris Cogeneration Llc-Brown Plant 230 kV Cir. 1	Reconductor line	\$15,872,000	68% (\$10,839,000)
	Total allocated to project		\$24,371,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.